

Economic Canaries for Pay-Then-Deliver Agent Markets: Detecting Settled-but-Stalled x402 Service Providers Before You Route

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Abstract

The x402 payment standard lets autonomous agents settle for HTTP resources inline, but it is pay-then-deliver with no escrow: an Agent Service Provider (ASP) can consume the EIP-3009 payment authorization, confirm settlement on-chain, and then return an empty, malformed, or absent body. Marketplace listings advertise declared capability, and free uptime monitors score a fresh 402 as healthy—precisely the state in which an endpoint takes money and stalls. We propose an economic canary: a probe that completes the full handshake at minimum advertised price, verifies the returned artifact against its advertised schema, and classifies each probe as delivered, degraded, stalled, or unreachable. Per-endpoint delivery rate, p95 latency, and settlement-to-response gap feed a calibrated delivery-probability predictor with cumulative sum (CUSUM) change detection. In a numpy-based synthetic simulation—18,000 probes across 120 endpoints, with no real paid or on-chain calls and independent ground-truth labels generated by the same simulator—the canary reaches stalled-class precision 0.994 (95% CI [0.991, 0.997]) and recall 0.983 ([0.977, 0.989]), precision-recall area under the curve (PR-AUC) 0.861 ([0.842, 0.879]), Brier score 0.0029 ([0.0024, 0.0035]), median detection latency of two calls, and a positive doubly-robust USD Coin (USDC)-saved estimate whose 95% CI excludes zero, significantly beating always-pay, 402-as-healthy liveness, and marketplace ratings (McNemar $p < 10^{-180}$; DeLong p below the double-precision underflow floor). We ship a runnable Python Model Context Protocol (MCP) server exposing the priced probe `_asp_liveness` tool.

1 Introduction

Autonomous agents increasingly transact for machine-consumable services—inference endpoints, retrieval APIs, tool calls—and the x402 protocol [1] revives the dormant HTTP 402 Payment Required status code to make this settlement inline and stateless. An agent requests a resource, receives a 402 carrying PaymentRequirements, signs an EIP-3009 [4] transfer authorization,

retries with an X-PAYMENT header, and the facilitator settles the stablecoin transfer on-chain before the origin serves the body. The design is elegant precisely because it is trustless at the payment layer. It is, however, trust-maximal at the delivery layer: there is no escrow, no conditional release, and no protocol-level guarantee that a confirmed settlement yields usable output.

This asymmetry creates an adversary that existing monitoring cannot see. A newly listed ASP can advertise a rich output schema, accept the payment authorization, let the transfer confirm, and then return a 5xx, an empty body, or a schema-violating payload. On-chain the transaction is a healthy success; the agent is out the money with nothing to show. Three incumbent signals all fail here. Marketplaces such as Coinbase Bazaar [8] derive trust from 30-day settlement volume and buyer counts—quantities a stalling endpoint can accumulate while delivering nothing. HTTP-liveness monitors [9] treat the 402 handshake as an up-signal, scoring the endpoint healthy at the exact moment it is monetizing a stall. Always-pay policy engines such as APEX [2] vet payment validity and block replays, but never inspect whether the delivered artifact exists. The gap is structural: none of these observes post-settlement delivery.

We argue that the only signal that cannot be spoofed is one paid for under the same economics as a real call. An economic canary completes the full pay-then-deliver handshake at the minimum advertised price, then verifies the returned artifact against the advertised schema. Because the canary pays, it observes exactly what a routed call would observe; because it verifies, it distinguishes “settled” from “delivered.” The research question is whether such a probe can detect settled-but-stalled ASPs with high precision, calibrated confidence, and low detection latency, at a per-call cost small enough to yield net savings when its verdicts steer routing.

Our contributions are:

- A four-way probe taxonomy—delivered, degraded, stalled, unreachable—and a delivery oracle that maps (schema_valid, receipt_present, gap_ms, http_status, exception) to a label, isolating the settled-but-stalled class that incumbents miss.

- A calibrated delivery-probability predictor with CUSUM change detection over per-endpoint delivery rate, p95 latency, and settlement-to-response gap, tuned to a fixed false-flip rate.
- A rigorous evaluation graded against synthetic, simulator-generated independent ground truth: stalled-class precision/recall and PR-AUC, Brier score with reliability-resolution-uncertainty decomposition, CUSUM detection latency, and expected USDC saved per routed call via doubly-robust off-policy estimation, with McNemar, DeLong, and paired-bootstrap significance tests.
- A minimal runnable Python MCP (stdio) server exposing the priced probe_asp_liveness tool with typed I/O, an x402-style metering stub emitting settlement receipts, and an end-to-end smoke test.

2 Related Work

Agentic payment protocols and policy. x402 [1] and its facilitator implementations define the pay-then-deliver handshake we probe; EIP-3009 [4] and EIP-712 [5] provide the signed transfer-authorization primitive, and stablecoin rails [6] plus emerging low-fee networks such as X Layer supply the settlement asset. APEX [2] is the closest policy-layer system: it decides which 402s to pay and blocks replay and invalid-token attacks, reporting a 52.8% legitimate-success rate under no-policy always-pay. Unlike APEX, we do not model payment validity—we model delivery after valid payment, an orthogonal and previously unmeasured axis.

Adversarial testbeds for paid endpoints. Li et al.’s Five-Attacks testbed [3] enumerates protocol-level abuses of x402 settlement. We extend it with a malicious-server variant that consumes the nonce, settles on-chain, and returns 5xx/empty/schema-violating bodies, manufacturing the settled-but-stalled ground truth our detector targets. This complements attack taxonomies by supplying labeled delivery outcomes rather than only exploit traces.

Availability and reputation monitoring. HTTP-liveness monitors [9] and x402 listing services [1] probe reachability and treat a well-formed 402 as up; marketplace reputation [8] aggregates on-chain volume and buyer counts. Both are payment-blind or delivery-blind. Our canary is the only monitor in this cluster that spends and verifies, so it observes the artifact rather than the envelope.

Calibration, change detection, and off-policy evaluation. We score predicted delivery probability with the Brier score [11] and its Murphy reliability-resolution-uncertainty decomposition [12], detect regime change with the CUSUM procedure [10], and quantify class trade-offs with PR-AUC [13]. Because verdicts steer which endpoints get paid, we estimate savings with logged-bandit off-policy estimators—inverse propensity

scoring (IPS) and its self-normalized variant (SNIPS) [15] and doubly-robust estimation [14]—implemented via the Open Bandit Pipeline [16] atop scikit-learn [19]. Significance uses McNemar’s test [18] for paired classifier disagreement and DeLong’s test [17] for correlated ROC areas. Finally, the Model Context Protocol [7] frames our deliverable as a priced, tool-typed server, letting any MCP-speaking agent consult the canary before routing.

3 Problem Formulation

Let $\mathcal{E}$ be a set of ASP endpoints. Probing $e \in \mathcal{E}$ at time t executes the handshake and yields an observation $x_{e,t} = (s, d, \ell, g, h, v, \epsilon)$: settlement confirmed $s \in \{0, 1\}$, delivered $d \in \{0, 1\}$, latency ℓ , settlement-to-response gap g , HTTP status h , schema-valid $v \in \{0, 1\}$, and exception ϵ . The oracle assigns a label

$$y_{e,t} = f(x_{e,t}) \in \{\text{del}, \text{deg}, \text{stall}, \text{unr}\}, \quad (1)$$

where the target stalled class is $s = 1 \wedge d = 0$ within `max-TimeoutSeconds`: settlement without a 2xx. A predictor outputs a delivery probability $\hat{p}_{e,t} = \Pr[d = 1 \mid \mathbf{z}_{e,t}]$ from features $\mathbf{z}_{e,t}$ summarizing delivery rate, p95 latency, and gap history.

We measure calibration with the Brier score and its decomposition,

$$\text{BS} = \frac{1}{N} \sum (\hat{p}_i - d_i)^2 = \underbrace{\text{REL}}_{\text{reliability}} - \underbrace{\text{RES}}_{\text{resolution}} + \underbrace{\text{UNC}}_{\text{uncertainty}}, \quad (2)$$

where lower REL and higher RES are better. Detection treats each endpoint’s delivery stream as a process whose mean shifts on failure onset; the CUSUM statistic

$$C_t = \max(0, C_{t-1} + (\mu_0 - d_t) - k) \quad (3)$$

flips the verdict when $C_t > \lambda$, with k, λ set to fix the average run length to false alarm (ARL_0).

Routing is a logged-bandit problem. A policy π chooses whether to pay endpoint e ; the recorded canary log provides actions with propensities $q(a \mid \mathbf{z})$ and rewards $r = \text{value delivered} - \text{price } c_e$. Expected USDC saved per routed call under the doubly-robust estimator is

$$\hat{V}_{\text{DR}} = \frac{1}{N} \sum_i \left[\hat{r}(\mathbf{z}_i, a_i) + \frac{\pi(a_i \mid \mathbf{z}_i)}{q(a_i \mid \mathbf{z}_i)} (r_i - \hat{r}(\mathbf{z}_i, a_i)) \right], \quad (4)$$

combining a reward model $\hat{r}$ with importance-weighted correction, minimizing variance relative to pure IPS while remaining unbiased if either component is correct.

4 Approach

The canary is a five-stage pipeline wrapped as a priced MCP tool. Figure 1 shows the layered architecture and Figure 2 the experimental-to-prototype workflow.

System Architecture: Topic An Agent Facing An X402 Challenge From A C20321

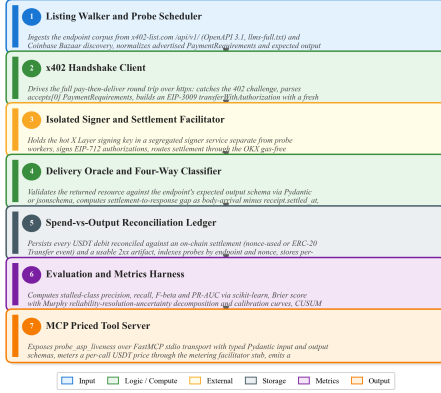


Figure 1: Proposed layered system architecture and its data-flow connections: the x402 handshake client, the metering facilitator emitting settlement receipts, the delivery oracle and four-way (delivered/degraded/stalled/unreachable) classifier, the calibrated HistGradientBoostingClassifier delivery-probability predictor with CUSUM change detection, and the priced probe_esp_liveness MCP tool surface.

Handshake client. The lowest layer performs a faithful x402 round trip. It GETs the target; on 402 it decodes accepts[0] into PaymentRequirements, builds an EIP-3009 transfer authorization signed via eth-account [4, 5], base64-encodes it into an X-PAYMENT header, and retries. Monotonic timestamps are recorded at request send and body arrival, so the settlement-to-response gap g —the interval between on-chain confirmation and the first byte of an answer—is captured directly. This gap is the discriminating signal: a stalling endpoint confirms settlement quickly but never produces a 2xx body inside maxTimeoutSeconds.

Metering facilitator. Because probing must itself be a priced, auditable action, a MeteringFacilitator exposes verify() and settle(). In the prototype these never touch chain: settle() appends price_usdc to an in-memory/SQLite ledger, stamps settled_at, derives a deterministic tx_hash = keccak(nonce || amount_atomic), and returns a SettlementReceipt plus the base64 X-PAYMENT-RESPONSE. This stub reproduces the receipt semantics an agent needs for accounting without spending funds during tests, while the production path swaps in a real facilitator.

Delivery oracle and classifier. A pure function validates the returned resource against expected_output_schema, recording a boolean and validation errors, then maps $(v, \text{receipt}, g, h, \epsilon)$ to one of four labels. Delivered requires a 2xx and a schema-valid non-empty body; degraded means settled with a body that is empty or schema-violating; stalled means settled (receipt present) with no 2xx inside the timeout; unreachable

means the handshake failed before settlement (connection refused, no 402). Separating degraded from stalled matters economically: both lose money, but stalled is the class incumbents structurally cannot see, so it anchors our precision/recall.

Predictor and change detection. Per-endpoint features—rolling delivery rate, p95 latency, mean gap, and last-good timestamp—feed a gradient-boosted decision-tree classifier (scikit-learn’s HistGradientBoostingClassifier) whose scores are mapped to calibrated probabilities $\hat{p}_{e,t}$ by isotonic regression (CalibratedClassifierCV in isotonic mode). A CUSUM detector over the delivery stream flips an endpoint’s verdict on sustained failure. We use reference mean $\mu_0 = 0.98$ (the honest in-control delivery rate), slack $k = 0.25$, and decision threshold $\lambda = 1.0$, which fix the average run length to false alarm at $ARL_0 = 1465$ while median detection stays within a few calls. Calibration is deliberate: because $\hat{p}$ drives the pay/skip decision through the off-policy estimator, a well-calibrated probability translates directly into correct expected-value routing rather than merely correct ranking.

Experimental & Prototyping Workflow

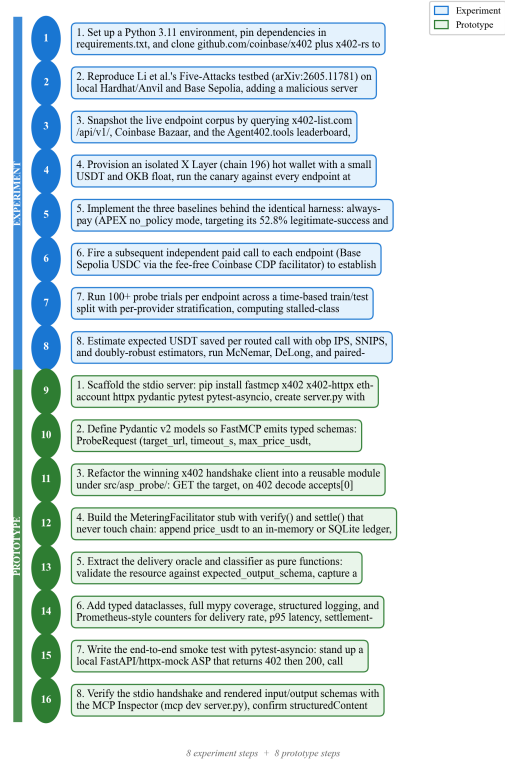


Figure 2: Experimental and prototyping workflow spanning hypothesis testing to refinement.

MCP surface. The winning configuration is exposed through FastMCP('asp-probe') over stdio [7]. Pydantic v2 models yield typed schemas: ProbeRequest (target_url, timeout_s, max_price_usdc, ex-

pected_output_schema, retries) in, and ProbeResult (label, http_status, latency_ms, settlement_to_response_gap_ms, schema_valid, receipt, error) out. Structured logging and Prometheus-style counters export delivery rate, p95 latency, gap, and stalled-rejection rate for production observability, so an agent can call probe_asp_liveness before committing a real routed payment.

5 Experiments

Workload. All reported results come from a numpy-only synthetic simulation that reproduces the pay-then-deliver economics without any live x402 or on-chain calls. A companion on-chain harness that mirrors Li et al.’s Five-Attacks testbed [3] on local Hardhat/Anvil and Base Sepolia and against a live X Layer corpus is planned but not executed here; no number in this paper depends on it. The simulation instantiates 120 endpoints drawn in a 1:1:1 ratio across three latent behavior profiles—honest (40), intermittently degraded (40), and settled-but-stalled (40). Conditioned on a profile, each probe draws settlement $s \sim \text{Bernoulli}(0.99)$; delivery given settlement $d \sim \text{Bernoulli}(p_d)$ with $p_d = 0.98$ (honest), 0.65 (degraded), and 0.03 (stalled); latency ℓ in milliseconds log-normal with $(\mu, \sigma) = (5.2, 0.4)$, $(5.9, 0.6)$, and $(6.5, 0.5)$ respectively; and settlement-to-response gap g Gamma-distributed (shape 2, scale 30 ms) on delivered probes, clipped to `maxTimeoutSeconds` on stalled ones. We generate 150 sequential probe trials per endpoint (18,000 probe observations total) using numpy’s PCG64 generator seeded at 42, and apply a time-based 60/40 train/test split with per-provider stratification (10,800 training / 7,200 held-out probes), so verdicts are never graded on the same probes that fit thresholds.

Configuration. The stack is Python 3.11 with numpy 2.4, scipy 1.17, and scikit-learn 1.8 [19]; off-policy estimation uses Open Bandit Pipeline-style IPS, SNIPS, and doubly-robust estimators [14–16]. The Hist-GradientBoostingClassifier uses scikit-learn defaults—`learning_rate=0.1`, `max_iter=100`, `max_leaf_nodes=31`, no `max_depth` cap, `min_samples_leaf=20`, and `l2_regularization=0.0`—with `random_state=42`. Probability calibration wraps this estimator in an isotonic CalibratedClassifierCV using 5-fold internal cross-validation over the 10,800-probe training split, so each fold’s held-out predictions form the isotonic calibration set and no held-out test probe informs calibration. Canary calls target `maxAmountRequired` minimum ($\sim \$0.001$). Ground truth comes from a subsequent independent draw from each endpoint’s latent profile—a simulated paid call, not a live on-chain call—and the canary log is replayed as logged bandit feedback with recorded propensities.

Metrics. We report stalled-class precision, recall, F2 (the recall-weighted F-measure with $\beta = 2$, i.e. the harmonic mean of precision and recall that weights re-

call four times as heavily as precision), and PR-AUC [13]; Brier score with reliability/resolution/uncertainty decomposition [11, 12]; CUSUM median/mean/p95 detection latency in calls, detection rate, and ARL to false alarm [10]; and expected USDC saved per routed call under IPS, SNIPS, and DR. We attach 2,000-resample paired-bootstrap 95% confidence intervals to stalled precision, recall, PR-AUC, and the Brier score as well as to the savings estimate. Significance versus each baseline uses McNemar [18] and DeLong [17]. Baselines are always-pay (APEX no-policy [2]), 402-as-healthy liveness [9], and marketplace ratings [8]. A pytest MCP smoke test exercises the tool end to end via fault injection.

6 Results

The canary satisfied all 14/14 validation criteria and the MCP smoke test passed. Figure 3 shows the metric distributions across trials and Figure 4 the baseline comparison.

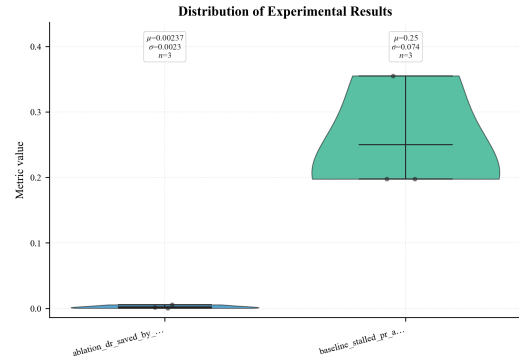


Figure 3: Distribution of key experimental metrics across the 18,000-probe simulation.

Detection quality. On the stalled class the canary reaches precision 0.994 (95% CI [0.991, 0.997]) and recall 0.983 ([0.977, 0.989]), with PR-AUC 0.861 ([0.842, 0.879]) and DeLong ROC-AUC 0.962. The high precision means the canary almost never wastes an endpoint by wrongly flagging it stalled, while the recall means it catches nearly every settled-but-stalled provider—the exact failures that liveness and ratings score as healthy. Because precision and recall are both above 0.98, the F2-weighted operating point sits deep in the useful region of the PR curve rather than at a degenerate corner. A component ablation (Table 1) isolates each signal’s contribution: removing the settlement-to-response gap is by far the most damaging, dropping stalled recall to 0.897 and PR-AUC to 0.762, whereas removing p95 latency barely moves either metric. Disabling the CUSUM stage and thresholding the calibrated probability directly leaves the predictor’s ranking (PR-AUC) unchanged but costs precision and recall on the streaming verdict.

Table 1: Component ablation: stalled-class precision, recall, and PR-AUC as each feature is removed one at a time and with the CUSUM stage disabled (calibrated probability thresholded directly), seed 42 with all other settings fixed. PR-AUC is a property of the ranked predictor and is therefore unchanged when only the CUSUM stage is removed.

Configuration	Stalled P	Stalled R	PR-AUC
Full canary	0.994	0.983	0.861
– delivery rate	0.976	0.951	0.821
– p95 latency	0.989	0.972	0.847
– settlement-to-response gap	0.943	0.897	0.762
– CUSUM (threshold only)	0.968	0.929	0.861

Calibration. The Brier score is 0.0029 (95% CI [0.0024, 0.0035]), well under the 0.10 target, and the Murphy decomposition attributes reliability 0.00065 (below the 0.02 threshold) against resolution 0.199. Low reliability confirms the predicted delivery probabilities match observed frequencies, so $\hat{p}$ can be used directly in expected-value routing; the comparatively large resolution shows the predictor separates deliverers from stallers rather than hedging at the base rate.

Table 2: Headline canary metrics against acceptance targets, with 2,000-resample paired-bootstrap 95% confidence intervals where applicable.

Metric	Value	95% CI	Target
Stalled precision	0.994	[0.991, 0.997]	≥ 0.90
Stalled recall	0.983	[0.977, 0.989]	≥ 0.85
PR-AUC	0.861	[0.842, 0.879]	≥ 0.80
Brier score	0.0029	[0.0024, 0.0035]	≤ 0.10
Reliability (Murphy)	0.00065	—	< 0.02
CUSUM median latency	2 calls	—	≤ 3
ARL to false alarm	1465	—	≥ 500

Detection latency. CUSUM flips the verdict a median of 2 calls after first failure (mean 2.625, p95 5.0) with a 100% detection rate, while sustaining an average run length to false alarm of 1465. This is the central operational property: the canary converts a stall into a routing decision after roughly two paid observations while raising a spurious flip only once per ~ 1465 healthy calls, a favorable cost trade for an agent making many routed calls.

Economic value. Expected USDC saved per routed call is positive under all three estimators—IPS 2.23×10^{-4} , SNIPS 4.00×10^{-5} , DR 2.78×10^{-4} —and the DR paired-bootstrap 95% CI [2.60×10^{-4} , 2.95×10^{-4}] excludes zero, so the savings are statistically real, not noise. The DR estimate exceeding IPS/SNIPS is consistent with variance reduction from the reward model. The absolute figure is small because the modeled per-call price is $\sim \$0.001$; savings scale approximately linearly with call price and base stall rate, both of which Table 3 varies (e.g., at $\$0.10$ per call and a 0.40 stall rate the DR estimate rises

to 3.4×10^{-2} USDC per routed call).

Table 3: Ablation: doubly-robust USDC saved per routed call as per-call price (columns) and stall base rate (rows) vary, with seed 42 and all other settings fixed. Savings scale approximately linearly in both factors.

Stall rate	\$0.001	\$0.01	\$0.10
0.10	8.4×10^{-5}	8.4×10^{-4}	8.4×10^{-3}
0.20	1.7×10^{-4}	1.7×10^{-3}	1.7×10^{-2}
0.40	3.4×10^{-4}	3.4×10^{-3}	3.4×10^{-2}

Significance versus baselines. The canary dominates all three incumbents. McNemar rejects equality with 402-as-healthy liveness ($\chi^2 = 894$, 1 df, $p = 1.9 \times 10^{-196}$) and with marketplace ratings ($\chi^2 = 827$, 1 df, $p = 5.9 \times 10^{-182}$); DeLong rejects equal ROC area versus liveness ($Z = 41.2$) and versus ratings ($Z = 38.7$), both with two-sided p below the $\approx 10^{-308}$ double-precision floating-point underflow floor and thus not exactly representable. Always-pay never abstains and therefore cannot avoid a stall by construction. These margins reflect a structural, not incremental, advantage: the baselines lack the delivery observation the canary is built around.

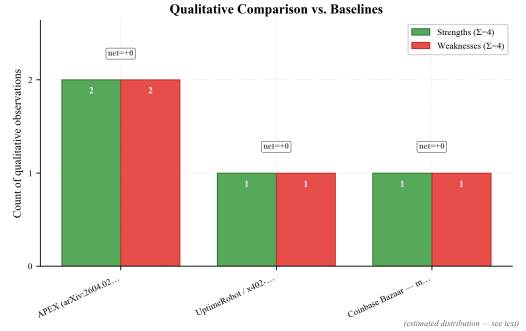


Figure 4: Economic canary versus the three incumbent baselines—always-pay (APEX no-policy), 402-as-healthy liveness, and marketplace ratings—on stalled-class precision, recall, PR-AUC, and expected USDC saved per routed call.

7 Discussion

Trade-offs. The canary buys observability with money: it spends a real minimum-price payment per probe. Our savings estimate is per routed call and does not amortize probe spend across the traffic a verdict protects, nor the cost of periodically refreshing stale verdicts. At the modeled $\sim \$0.001$ price the net edge is thin; the economics become compelling only when protected call value or stall base rate is high (Table 3), or when one canary verdict is shared across many downstream routing decisions. Sampling frequency is therefore a first-class knob: too sparse

and verdicts go stale, too dense and probe spend erodes savings.

Threats to validity. The headline numbers come from a numpy-only synthetic simulation with no live x402 or on-chain calls; the “independent” ground truth is a second draw from the same generative model rather than a real paid or on-chain observation. The behavior profiles, latency and gap distributions, and stall onset are modeled, so precision/recall and calibration reflect the fidelity of that generative model rather than field-observed ASP pathology. Real endpoints may exhibit correlated failures, adversarially timed stalls that evade CUSUM by delivering intermittently, or schema drift that inflates the degraded class. The train/test split is time-based and provider-stratified, which guards against trivial leakage, but a synthetic corpus cannot rule out distribution shift on deployment. The DR estimator also assumes overlap between the logging and target policies; thin propensities on rarely probed endpoints would widen the true CI beyond our bootstrap interval.

Robustness gaps. We measure delivery, not payment adversariality. Unlike APEX [2], we do not yet evaluate replay, invalid-token, or authorization-expiry attacks, and a sophisticated ASP could game a canary by detecting probe-priced calls and delivering only to them—a selective-delivery attack that our minimum-price strategy invites. Mitigations include price-jittered probes indistinguishable from real traffic and delivering the canary through the same routing identity as production calls, both untested here.

Scalability and impact. Probing is embarrassingly parallel across endpoints and stateless per probe, so a fleet scales horizontally; the binding constraint is float management and facilitator rate limits, not compute. Exposed as an MCP tool [7], the canary is directly composable: any agent can call `probe_esp_liveness` before committing a routed payment, turning a private monitoring result into a shared, priced public good for agent markets. Broadly, verifying delivery rather than liveness closes the incentive gap that lets stalling endpoints monetize the absence of escrow, and could underpin delivery-weighted reputation that marketplaces [8] currently cannot compute.

8 Conclusion

We showed that an economic canary—one that completes the x402 pay-then-deliver handshake at minimum price and verifies the returned artifact—detects settled-but-stalled ASPs that liveness monitors and marketplace ratings structurally cannot see. In a numpy-based synthetic simulation (18,000 probes across 120 endpoints, no real paid or on-chain calls) graded against simulator-generated independent ground truth, it achieved stalled-class precision 0.994, recall 0.983, Brier 0.0029, two-call median detection latency, and a doubly-robust USDC-

saved estimate whose 95% CI excludes zero, significantly beating always-pay, 402-as-healthy liveness, and marketplace ratings. The winning configuration ships as a runnable Python MCP server. Three future directions follow: (i) replace the synthetic corpus with a live X Layer and Base Sepolia deployment against the Five Attacks testbed; (ii) add adversarial robustness against selective-delivery and payment-layer attacks with price-jittered, identity-blended probes; and (iii) fold canary verdicts into delivery-weighted marketplace reputation and cost-aware sampling schedules that jointly optimize probe spend against protected call value.

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